Consumption of a boiled Greek type of coffee is associated with improved endothelial function: the Ikaria study.

OBJECTIVE: The association of coffee consumption with cardiovascular disease remains controversial. Endothelial function is associated with cardiovascular risk. We examined the association between chronic coffee consumption and endothelium function in elderly inhabitants of the island of Ikaria. METHODS: The analysis was conducted on 142 elderly subjects (aged 66-91 years) of the Ikaria Study. Endothelial function was evaluated by ultrasound measurement of flow-mediated dilation (FMD). Coffee consumption was evaluated based on a food frequency questionnaire and was categorized as 'low' (< 200 ml/day), 'moderate' (200-450 ml/day), or 'high' (> 450 ml/day). RESULTS: From the subjects included in the study, 87% consumed a boiled Greek type of coffee. Moreover, 40% had a 'low', 48% a 'moderate' and 13% a 'high' daily coffee consumption. There was a linear increase in FMD according to coffee consumption ('low': 4.33 ± 2.51% vs 'moderate': 5.39 ± 3.09% vs 'high': 6.47 ± 2.72%; $p = 0.032$). Moreover, subjects consuming mainly a boiled Greek type of coffee had a significantly higher FMD compared with those consuming other types of coffee beverages ($p = 0.035$). CONCLUSIONS: Chronic coffee consumption is associated with improved endothelial function in elderly subjects, providing a new connection between nutrition and vascular health.